

D043 restrained apparatus

Physical page 1 / SEALS repository wrapper

Container role. Repository provenance and copy matter. The page is preserved byte-for-byte in the source container and excluded in full from both reader bodies.

Directly logged wrapper matter.

- Title: *Division Algebras and the Hausdorff-Banach-Tarski Paradox*.
- Repository names: Deligne, Pierre / Sullivan, Dennis. These names confirm joint authorship; the PDF catalog's Deligne-only Author field is false by omission.
- Journal/volume: *L'Enseignement Mathématique*, 29 (1983), issue 1-2.
- Persistent-link display and annotation: <http://dx.doi.org/10.5169/seals-52976>. This is source provenance only, not a new release identifier.
- Creation/use/custody matter: created May 23, 2012; SEALS access terms; Swiss academic-library consortium custody; ETH-Bibliothek address and contact matter.

Reader disposition. EXCLUDE_ENTIRE_WRAPPER_FROM_BOTH_READERS.

Authority identity. 401,568 bytes; seven physical pages; SHA-256:

6312B2B7281723208434455BB590018ECD3542501EF603890B3F1D6283E4A4F0

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Physical page 2 / article page 1 / printed 145

Preserved in both readers. Publication line; full title; exact joint byline; opening partition notation $E, X, \bar{X}, Y, \bar{Y}$; the two displayed reassembly equations; the sole partition-and-reassembly diagram; and the italic reassembly statement continuing to printed 146.

Diagram authority. The inherited TikZ approximation is rejected. The reader uses the registered native one-bit authority crop at half-open bounds [120,2760,2880,4090) without resampling, filtering, tracing, cleanup, or redraw.

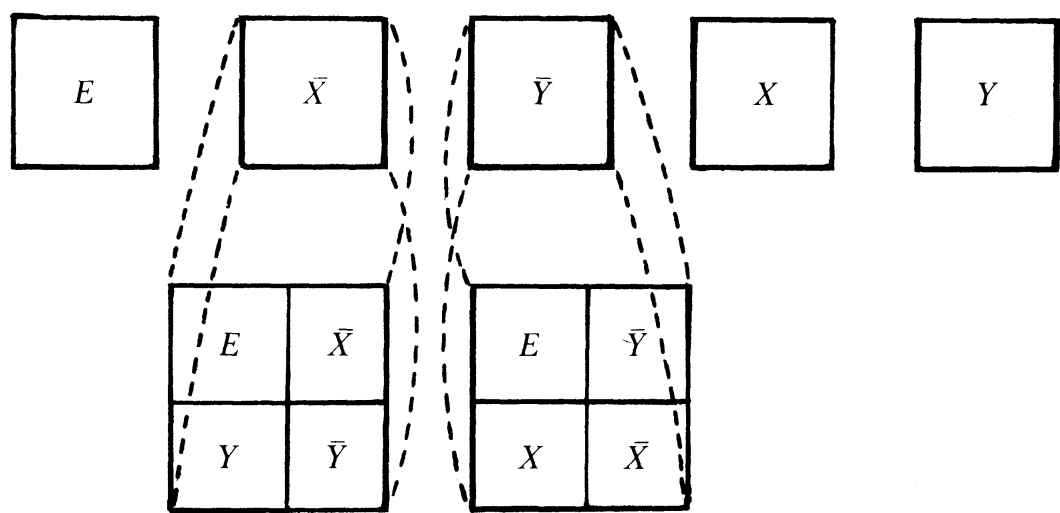


Diagram SHA-256.

CCA25D5EC3E328B310B71A2B657326A69F8280A56827AA1A471A7FDFFAD5249C

Copy matter excluded from reader prose. Bottom publisher/footer line and signature number; scan-edge artifacts. The publication line at the top is retained as required source matter.

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Physical page 3 / article page 2 / printed 146

Preserved in both readers. Continuation from printed 145; Dekker/Mycielski question; odd-dimensional restriction; Lefschetz formula; unnumbered THEOREM; three separately printed *Remark.* paragraphs; and the opening of *Proof.*

Structural replay.

- The theorem remains unnumbered. No automatic theorem counter is introduced.
- Exactly three printed Remarks are retained, without generated numbering or labels.
- Every exponent, inverse, bar, tensor subscript, arrow, matrix block, and group symbol is replayed at its page address.
- The proof continues at the physical-page boundary after $SU(n)$.

Copy matter excluded from reader prose. Folio 146, running head “P. DELIGNE AND D. SULLIVAN”, and scan-edge artifacts.

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Physical page 4 / article page 3 / printed 147

Preserved in both readers. Density of Γ in $U(n)$; skew-adjoint elements; Cayley transform $t \mapsto (t - 1)/(t + 1)$; Tits free-subgroup conclusion; the Brauer norm criterion; Hermitian adjunction argument; and complete APPENDIX A.

Page-boundary control. The continuation from printed 146 begins “of $U(n)$ is not trivial” and the page terminates at the end of Appendix A. No material from Appendix B is moved backward into this page.

Copy matter excluded from reader prose. Folio 147, running head “DIVISION ALGEBRAS”, and scan-edge artifacts.

P01 coverage. Wrapper disposition plus printed 145-147 are complete in diplomatic English, standalone English, and page-addressed apparatus. Workflow cursor advances to P02.

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Physical page 5 / article page 4 / printed 148

Preserved in both readers. Complete opening page of APPENDIX B: the separable quadratic extension K/k ; conjugate modules and barred scalars; D' , $\bar{D}'$, V' , $\bar{V}'$; the fixed Hermitian subspace W' ; the algebra D'_1 ; involutions of the second kind; and the Severi-Brauer variety criterion.

Symbol-level controls.

- Bars and primes remain distinct throughout.
- Tensor bases k , k' , K , and K' are preserved at each occurrence.
- The map $v \otimes \bar{w} \mapsto w \otimes \bar{v}$ and the fixed-subspace construction are not paraphrased.

Copy matter excluded from reader prose. Folio 148, running head “P. DELIGNE AND D. SULLIVAN”, and scan-edge artifacts.

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Physical page 6 / article page 5 / printed 149

Preserved in both readers. Conclusion of Appendix B; norm $N_{K/k}(\text{cl}(D))$; homothety by $\lambda \in K'^*$; complete APPENDIX C; cyclotomic algebra generators and relations; involution formulas; and the two numerical examples.

Direct authority resolution of packet contradiction. The controlling page prints

$$\text{Fix } a, b \in k^*.$$

There is no prime on k . The P02 literal prompt and inherited hazard ledger assert k'^* , but that assertion conflicts with the controlling authority and is rejected.

APPENDIX C

For $n \geq 3$, examples can be obtained as follows: take $k' = \mathbf{Q}[\zeta]$, with $\zeta = \exp(2\pi i/n)$, and $k = k' \cap \mathbf{R}$. Fix $a, b \in k^*$ and let D be the k' -algebra generated by X, Y , subject to

$$X^n = a \qquad Y^n = b$$

Evidence-crop SHA-256.

5CCA798ED2352C04CDC4F565CD38F1222A80BAFD0F0E44158160760E30CB7F38

Copy matter excluded from reader prose. Folio 149, running head “DIVISION ALGEBRAS”, and scan-edge artifacts.

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Physical page 7 / article page 6 / printed 150

Preserved in both readers. All five bibliography entries; the printed spelling “Mycielkski, Jan.”; (*Reçu le 8 mars 1982*); Pierre Deligne; Dennis Sullivan; the joint IHES address exactly as printed; blank terminal remainder; and physical EOF.

BIBLIOGRAPHY

- [A] ALBERT, A. *Structure of algebras*. A.M.S. Colloquium Publ. 24 (1939).
- [Sch] SCHARLAU, W. Zur Existenz von Involutionen auf einfachen Algebren. *Math. Zeitschr.* 145 (1975), 29-32.
- [Myc] MYCIELKSKI, Jan. Can one solve equations in groups? *Amer. Math. Monthly* 84 (1977), 723-726.
- [Tits] TITS, J. Free subgroups in linear groups. *Journal of Algebra* 20 (1972), 250-270.
- [D] DEKKER, Th. J. Decomposition of sets and spaces I, II. *Indig. Math.* 18 (1956), 581-595, and 19 (1957), 104-107.

(*Reçu le 8 mars 1982*)

Pierre Deligne
Dennis Sullivan

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Bibliography crop SHA-256. B0AB79209F2B7C49327E0098FD2754DC41A1DDC9F7700CE8908331FC9E67B28D

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Terminal disposition. The printed address reads “Institut des Hautes Etudes Scientifiques” without an accent on *Etudes*. The large blank remainder is retained as terminal-page evidence in apparatus and logged through physical EOF; it is not converted into reader prose.

Copy matter excluded from reader prose. Folio 150, running head “P. DELIGNE AND D. SULLIVAN”, and scan-edge artifacts.